Manage event destinations

ONTAP 9.13.1 REST API reference

NetApp
February 13, 2024

This PDF was generated from https://docs.netapp.com/us-en/ontap-restapi-9131/ontap/support_ems_destinations_endpoint_overview.html on February 13, 2024. Always check docs.netapp.com for the latest.

Table of Contents

Manage event destinations . . . 1
Support EMS destinations endpoint overview . . . 1
Retrieve a collection of event destinations . . . 5
Create an event destination . . . 17

Manage event destinations

Support EMS destinations endpoint overview

Overview

Manages the list of destinations. A destination is defined by a type and a place to which an event's information is transmitted.

Destination Types

An EMS destination is defined by a single type, which is one of the following:

- email
- syslog
- rest_api
- snmp

Email

The 'email' type allows you to define a mailbox where information about an observed event is sent by SMTP. Enter the address in the destination field in a valid format. For example: administrator@mycompany.com

Syslog

The 'syslog' type allows you to specify a remote syslog server that can receive information about an observed event. Enter the hostname or IP address in the destination field. For example: syslog.mycompany.com, 192.168.1.1

You can optionally specify the port and transport protocol to be used. The supported transport protocols are:

- udp_unencrypted
- tcp_unencrypted
- tcp_encrypted

#

If transport is specified and the port is not specified, the default port for each transport is automatically selected. The default port for udp_unencrypted is 514, for tcp_unencrypted 601, and for tcp_encrypted 6514. The default transport protocol is udp_unencrypted.

#

The message format to be used can also be optionally specified. The supported message formats are:

- legacy_netapp (format: <PRIVAL>TIMESTAMP [HOSTNAME:Event-name:Event-severity]: MSG)
- rfc_5424 (format: <PRIVAL>VERSION TIMESTAMP HOSTNAME Event-source - Event-name - MSG)

#

If the default timestamp or hostname formats in the message format needs to be overridden, use timestamp_override and hostname_override properties.

The supported timestamp formats are:

- no_override (timestamp format based on the syslog.format.message property i.e. rfc_3164 if syslog.format.message is legacy_netapp, iso_8601_local_time if syslog.format.message is rfc_5424)
- rfc_3164 (format: Mmm dd hh:mm:ss)
- iso_8601_local_time (format: YYYY-MM-DDThh:mm:ss+/-hh:mm)
- iso_8601_utc (format: YYYY-MM-DDThh:mm:ssZ)

#

The supported hostname formats are:

- no_override (hostname format based on the syslog.format.message property i.e. fqdn if syslog.format.message is rfc_5424, hostname_only if syslog.format.message is legacy_netapp)
- fqdn (Fully Qualified Domain Name)
- hostname_only

Rest API

The 'rest_api' type allows you to define a URL where information about an observed event is sent using the REST protocol. Enter the URL in the destination field. The URL must contain a valid transmission schema which can be one of the following:

- http
- https

#

Using the *https* schema, you can configure a client-side certificate for mutual authentication. A client-side certificate is specified by the `ca` and `serial_number` fields in the `certificate` object. The `name` field of the `certificate` object is read-only and cannot be used to configure a client-side certificate.

For example: http://rest.mycompany.com, https://192.168.1.1

SNMP

The 'snmp' type describes addresses where information about an observed event is sent using SNMP traps. The system defines a default instance of this type and it is restricted to read-only. This type has the following limitations:

- Cannot create new destinations of type 'snmp'
- Cannot modify the default 'snmp' destination

SNMP trap host details need to be configured through one of the following:

Type	Command / API
CLI	'system snmp traphost'

Type	Command / API
ZAPI	'snmp-traphost-add' / 'snmp-traphost-delete'

Examples

Retrieving the list of active destinations

```
# The API:
GET /api/support/ems/destinations

# The call:
curl -X GET "https://<mgmt-ip>/api/support/ems/destinations" -H "accept:
application/hal+json"

# The response:
200 OK

# JSON Body
{
"records": [
  {
    "name": "snmp-traphost",
    "_links": {
      "self": {
        "href": "/api/support/ems/destinations/snmp-traphost"
      }
    }
  }
],
"num_records": 1,
"_links": {
  "self": {
    "href": "/api/support/ems/destinations"
  }
}
}
```

Creating a new 'email' destination

```
# The API:
POST /support/ems/destinations

# The call:
curl -X POST "https://<mgmt-ip>/api/support/ems/destinations" -H "accept:
application/hal+json" -H "Content-Type: application/json" -d
"@test_ems_destinations_post.txt"
test_ems_destinations_post.txt(body):

# JSON Body
{
"name": "Technician_Email",
"type": "email",
"destination": "technician@mycompany.com",
"filters": [
  { "name" : "critical-wafl" }
]
}

# The response:
201 Created
```

Creating a new 'syslog' destination

```
# The API:
POST /support/ems/destinations

# The call:
curl -X POST "https://<mgmt-ip>/api/support/ems/destinations" -H "accept:
application/hal+json" -H "Content-Type: application/json" -d
"@test_ems_destinations_syslog_post.txt"
test_ems_destinations_syslog_post.txt(body):

# JSON Body
{
"name": "Syslog_dest",
"type": "syslog",
"destination": "syslog.mycompany.com",
"syslog": {
  "port": 601,
  "transport": "tcp_unencrypted",
  "format": {
    "message": "rfc_5424",
    "hostname_override": "fqdn",
    "timestamp_override": "iso_8601_local_time"
  }
},
"filters": [
  { "name" : "critical-wafl" }
]
}

# The response:
201 Created
```

Retrieve a collection of event destinations

GET `/support/ems/destinations`

Introduced In: 9.6

Retrieves a collection of event destinations.

Related ONTAP commands

- `event notification destination show`
- `event notification show`

Parameters

Name	Type	In	Required	Description
type	string	query	False	Filter by type
connectivity.errors.node.uuid	string	query	False	Filter by connectivity.errors.node.uuid • Introduced in: 9.11
connectivity.errors.node.name	string	query	False	Filter by connectivity.errors.node.name • Introduced in: 9.11
connectivity.errors.message.code	string	query	False	Filter by connectivity.errors.message.code • Introduced in: 9.11
connectivity.errors.message.message	string	query	False	Filter by connectivity.errors.message.message • Introduced in: 9.11
connectivity.errors.message.arguments.message	string	query	False	Filter by connectivity.errors.message.arguments.message • Introduced in: 9.11
connectivity.errors.message.arguments.code	string	query	False	Filter by connectivity.errors.message.arguments.code • Introduced in: 9.11

Name	Type	In	Required	Description
connectivity.state	string	query	False	Filter by connectivity.state • Introduced in: 9.11
filters.name	string	query	False	Filter by filters.name
access_control_role.name	string	query	False	Filter by access_control_role.name • Introduced in: 9.13
certificate.ca	string	query	False	Filter by certificate.ca • maxLength: 256 • minLength: 1
certificate.name	string	query	False	Filter by certificate.name • Introduced in: 9.11
certificate.serial_number	string	query	False	Filter by certificate.serial_number • maxLength: 40 • minLength: 1
syslog.port	integer	query	False	Filter by syslog.port • Introduced in: 9.12
syslog.transport	string	query	False	Filter by syslog.transport • Introduced in: 9.12

Name	Type	In	Required	Description
syslog.format.timestamp_override	string	query	False	Filter by syslog.format.timestamp_override • Introduced in: 9.12
syslog.format.message	string	query	False	Filter by syslog.format.message • Introduced in: 9.12
syslog.format.hostname_override	string	query	False	Filter by syslog.format.hostname_override • Introduced in: 9.12
name	string	query	False	Filter by name
destination	string	query	False	Filter by destination
system_defined	boolean	query	False	Filter by system_defined • Introduced in: 9.10
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Default value: 1 • Max value: 120 • Min value: 0
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
num_records	integer	Number of records
records	array[records]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "access_control_role": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "admin"
    },
    "certificate": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "ca": "VeriSign",
      "name": "cert1",
      "serial_number": 1234567890
    },
    "connectivity": {
      "errors": {
        "message": {
          "arguments": {
            "code": "string",
            "message": "string"
          },
          "code": "4",
          "message": "entry doesn't exist"
        },
```

```
        "node": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "name": "node1",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "state": "fail"
    },
    "destination": "<a href="
mailto:administrator@mycompany.com">administrator@mycompany.com</a>",
    "filters": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "important-events"
    },
    "name": "Admin_Email",
    "syslog": {
      "format": {
        "hostname_override": "no_override",
        "message": "legacy_netapp",
        "timestamp_override": "no_override"
      },
      "port": 514,
      "transport": "udp_unencrypted"
    },
    "system_defined": 1,
    "type": "email"
  }
}
```


Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
next	href	
self	href	

_links

Name	Type	Description
self	href	

access_control_role

Indicates the access control role that created the event destination and is used to control access to the destination based on role-based access control (RBAC) rules. If created by the 'admin' user, the field is unset.

Name	Type	Description
_links	_links	
name	string	Role name

certificate

Specifies the client-side certificate used by the ONTAP system when mutual authentication is required. This object is only applicable for **rest_api** type destinations. Both the `ca` and `serial_number` fields must be specified when configuring a certificate in a PATCH or POST request. The `name` field is read-only and cannot be used to configure a client-side certificate.

Name	Type	Description
_links	_links	
ca	string	Client certificate issuing CA
name	string	Certificate name
serial_number	string	Client certificate serial number

arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

message

Name	Type	Description
arguments	array[arguments]	Message arguments
code	string	Unique message code.
message	string	User message.

node

Name	Type	Description
_links	_links	
name	string	
uuid	string	

errors

Error object included in the event of connectivity failure.

Name	Type	Description
message	message	
node	node	

connectivity

Name	Type	Description
errors	array[errors]	A list of errors encountered during connectivity checks.
state	string	Current connectivity state.

filters

Name	Type	Description
_links	_links	
name	string	

format

Name	Type	Description
hostname_override	string	Syslog Hostname Format Override. The supported hostname formats are no_override (hostname format based on the syslog.format.message property i.e. fqdn if syslog.format.message is rfc_5424, hostname_only if syslog.format.message is legacy_netapp), fqdn (Fully Qualified Domain Name) and hostname_only.
message	string	Syslog Message Format. The supported message formats are legacy_netapp (format: <PRIVAL>TIMESTAMP [HOSTNAME:Event-name:Event-severity]: MSG) and rfc_5424 (format: <PRIVAL>VERSION TIMESTAMP HOSTNAME Event-source - Event-name - MSG).
timestamp_override	string	Syslog Timestamp Format Override. The supported timestamp formats are no_override (timestamp format based on the syslog.format.message property i.e. rfc_3164 if syslog.format.message is legacy_netapp, iso_8601_local_time if syslog.format.message is rfc_5424), rfc_3164 (format: Mmm dd hh:mm:ss), iso_8601_local_time (format: YYYY-MM-DDThh:mm:ss+/-hh:mm) and iso_8601_utc (format: YYYY-MM-DDThh:mm:ssZ).

syslog

Name	Type	Description
format	format	

Name	Type	Description
port	integer	Syslog Port.
transport	string	Syslog Transport Protocol.

records

Name	Type	Description
_links	_links	
access_control_role	access_control_role	Indicates the access control role that created the event destination and is used to control access to the destination based on role-based access control (RBAC) rules. If created by the 'admin' user, the field is unset.
certificate	certificate	Specifies the client-side certificate used by the ONTAP system when mutual authentication is required. This object is only applicable for **rest_api** type destinations. Both the `ca` and `serial_number` fields must be specified when configuring a certificate in a PATCH or POST request. The `name` field is read-only and cannot be used to configure a client-side certificate. • Introduced in: 9.6
connectivity	connectivity	
destination	string	Event destination
filters	array[filters]	
name	string	Destination name. Valid in POST.
syslog	syslog	
system_defined	boolean	Flag indicating system-defined destinations.
type	string	Type of destination. Valid in POST.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Create an event destination

POST `/support/ems/destinations`

Introduced In: 9.6

Creates an event destination.

Required properties

- `name` - String that uniquely identifies the destination.
- `type` - Type of destination that is to be created.
- `destination` - String that identifies the destination. The contents of this property changes depending on type.

Recommended optional properties

- `filters.name` - List of filter names that should direct to this destination.
- `certificate` - When specifying a rest api destination, a client certificate can be provided.
- `syslog` - When specifying a syslog destination, a port, transport protocol, message format, timestamp format and hostname format can be provided.

Related ONTAP commands

- `event notification destination create`

- `event notification create`

Parameters

Name	Type	In	Required	Description
return_records	boolean	query	False	The default is false. If set to true, the records are returned. • Default value:

Request Body

Name	Type	Description
_links	_links	
access_control_role	access_control_role	Indicates the access control role that created the event destination and is used to control access to the destination based on role-based access control (RBAC) rules. If created by the 'admin' user, the field is unset.
certificate	certificate	Specifies the client-side certificate used by the ONTAP system when mutual authentication is required. This object is only applicable for **rest_api** type destinations. Both the `ca` and `serial_number` fields must be specified when configuring a certificate in a PATCH or POST request. The `name` field is read-only and cannot be used to configure a client-side certificate. • Introduced in: 9.6
connectivity	connectivity	
destination	string	Event destination
filters	array[filters]	
name	string	Destination name. Valid in POST.
syslog	syslog	

Name	Type	Description
system_defined	boolean	Flag indicating system-defined destinations.
type	string	Type of destination. Valid in POST.

Example request


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "access_control_role": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "admin"
  },
  "certificate": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "ca": "VeriSign",
    "name": "cert1",
    "serial_number": 1234567890
  },
  "connectivity": {
    "errors": {
      "message": {
        "arguments": {
          "code": "string",
          "message": "string"
        },
        "code": "4",
        "message": "entry doesn't exist"
      },
      "node": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "node1",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
```

```
    "state": "fail"
  },
  "destination": "<a href="
mailto:administrator@mycompany.com">administrator@mycompany.com</a>",
  "filters": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "important-events"
  },
  "name": "Admin_Email",
  "syslog": {
    "format": {
      "hostname_override": "no_override",
      "message": "legacy_netapp",
      "timestamp_override": "no_override"
    },
    "port": 514,
    "transport": "udp_unencrypted"
  },
  "system_defined": 1,
  "type": "email"
}
```


Response

```
Status: 201, Created
```

Name	Type	Description
_links	_links	
num_records	integer	Number of records
records	array[records]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "access_control_role": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "admin"
    },
    "certificate": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "ca": "VeriSign",
      "name": "cert1",
      "serial_number": 1234567890
    },
    "connectivity": {
      "errors": {
        "message": {
          "arguments": {
            "code": "string",
            "message": "string"
          },
          "code": "4",
          "message": "entry doesn't exist"
        },
```

```
        "node": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "name": "node1",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "state": "fail"
    },
    "destination": "<a href="
mailto:administrator@mycompany.com">administrator@mycompany.com</a>",
    "filters": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "important-events"
    },
    "name": "Admin_Email",
    "syslog": {
      "format": {
        "hostname_override": "no_override",
        "message": "legacy_netapp",
        "timestamp_override": "no_override"
      },
      "port": 514,
      "transport": "udp_unencrypted"
    },
    "system_defined": 1,
    "type": "email"
  }
}
```


Headers

Name	Description	Type
Location	Useful for tracking the resource location	string

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
983104	The syslog destination provided is invalid
983107	A provided filter does not exist
983116	The number of notifications has reached maximum capacity
983117	The number of destinations has reached maximum capacity
983129	The rest-api destination provided must contain a valid scheme, such as http// or https//
983130	The rest-api destination provided contains an invalid URL
983131	The rest-api destination provided contains an invalid IPv6 URL
983144	The security certificate information provided is incomplete. Provide the certificate and serial number
983145	The rest-api destination provided has an 'http://' scheme. It is invalid to provide certificate information
983149	New SNMP destinations cannot be created
983152	Default destinations cannot be modified or removed
983153	The security certificate provided does not exist
983154	The necessary private key is not installed on the system
983170	Connectivity check is not supported for the specified destination type on the node
983171	Connectivity check failed for the specified destination on the node
983176	Connectivity check is only supported for TCP-based syslog destinations
983179	The value for the destination field cannot be empty
983180	The destination name provided cannot be empty
983181	The destination name provided cannot contain spaces

Error Code	Description
983182	The destination name provided is invalid. The destination name must contain between 2 and 64 characters and start and end with an alphanumeric symbol or _(underscore). The allowed special characters are _(underscore) and -(hyphen)
983184	A provided property cannot be configured on the requested destination type
983200	Access control role compatibility issue with provided filters

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

access_control_role

Indicates the access control role that created the event destination and is used to control access to the destination based on role-based access control (RBAC) rules. If created by the 'admin' user, the field is unset.

Name	Type	Description
_links	_links	
name	string	Role name

certificate

Specifies the client-side certificate used by the ONTAP system when mutual authentication is required. This object is only applicable for **rest_api** type destinations. Both the `ca` and `serial_number` fields must be specified when configuring a certificate in a PATCH or POST request. The `name` field is read-only and cannot be used to configure a client-side certificate.

Name	Type	Description
_links	_links	
ca	string	Client certificate issuing CA
name	string	Certificate name
serial_number	string	Client certificate serial number

arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

message

Name	Type	Description
arguments	array[arguments]	Message arguments
code	string	Unique message code.
message	string	User message.

node

Name	Type	Description
_links	_links	
name	string	
uuid	string	

errors

Error object included in the event of connectivity failure.

Name	Type	Description
message	message	
node	node	

connectivity

Name	Type	Description
errors	array[errors]	A list of errors encountered during connectivity checks.
state	string	Current connectivity state.

filters

Name	Type	Description
_links	_links	
name	string	

format

Name	Type	Description
hostname_override	string	Syslog Hostname Format Override. The supported hostname formats are no_override (hostname format based on the syslog.format.message property i.e. fqdn if syslog.format.message is rfc_5424, hostname_only if syslog.format.message is legacy_netapp), fqdn (Fully Qualified Domain Name) and hostname_only.
message	string	Syslog Message Format. The supported message formats are legacy_netapp (format: <PRIVAL>TIMESTAMP [HOSTNAME:Event-name:Event-severity]: MSG) and rfc_5424 (format: <PRIVAL>VERSION TIMESTAMP HOSTNAME Event-source - Event-name - MSG).
timestamp_override	string	Syslog Timestamp Format Override. The supported timestamp formats are no_override (timestamp format based on the syslog.format.message property i.e. rfc_3164 if syslog.format.message is legacy_netapp, iso_8601_local_time if syslog.format.message is rfc_5424), rfc_3164 (format: Mmm dd hh:mm:ss), iso_8601_local_time (format: YYYY-MM-DDThh:mm:ss+/-hh:mm) and iso_8601_utc (format: YYYY-MM-DDThh:mm:ssZ).

syslog

Name	Type	Description
format	format	
port	integer	Syslog Port.

Name	Type	Description
transport	string	Syslog Transport Protocol.

ems_destination

Name	Type	Description
_links	_links	
access_control_role	access_control_role	Indicates the access control role that created the event destination and is used to control access to the destination based on role-based access control (RBAC) rules. If created by the 'admin' user, the field is unset.
certificate	certificate	Specifies the client-side certificate used by the ONTAP system when mutual authentication is required. This object is only applicable for **rest_api** type destinations. Both the `ca` and `serial_number` fields must be specified when configuring a certificate in a PATCH or POST request. The `name` field is read-only and cannot be used to configure a client-side certificate. • Introduced in: 9.6
connectivity	connectivity	
destination	string	Event destination
filters	array[filters]	
name	string	Destination name. Valid in POST.
syslog	syslog	
system_defined	boolean	Flag indicating system-defined destinations.
type	string	Type of destination. Valid in POST.

_links

Name	Type	Description
next	href	
self	href	

records

Name	Type	Description
_links	_links	
access_control_role	access_control_role	Indicates the access control role that created the event destination and is used to control access to the destination based on role-based access control (RBAC) rules. If created by the 'admin' user, the field is unset.
certificate	certificate	Specifies the client-side certificate used by the ONTAP system when mutual authentication is required. This object is only applicable for **rest_api** type destinations. Both the `ca` and `serial_number` fields must be specified when configuring a certificate in a PATCH or POST request. The `name` field is read-only and cannot be used to configure a client-side certificate. • Introduced in: 9.6
connectivity	connectivity	
destination	string	Event destination
filters	array[filters]	
name	string	Destination name. Valid in POST.
syslog	syslog	
system_defined	boolean	Flag indicating system-defined destinations.
type	string	Type of destination. Valid in POST.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Copyright information

Copyright © 2024 NetApp, Inc. All Rights Reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means—graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system—without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

LIMITED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (b)(3) of the Rights in Technical Data -Noncommercial Items at DFARS 252.227-7013 (FEB 2014) and FAR 52.227-19 (DEC 2007).

Data contained herein pertains to a commercial product and/or commercial service (as defined in FAR 2.101) and is proprietary to NetApp, Inc. All NetApp technical data and computer software provided under this Agreement is commercial in nature and developed solely at private expense. The U.S. Government has a non-exclusive, non-transferrable, nonsublicensable, worldwide, limited irrevocable license to use the Data only in connection with and in support of the U.S. Government contract under which the Data was delivered. Except as provided herein, the Data may not be used, disclosed, reproduced, modified, performed, or displayed without the prior written approval of NetApp, Inc. United States Government license rights for the Department of Defense are limited to those rights identified in DFARS clause 252.227-7015(b) (FEB 2014).

Trademark information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.